# EDS Lab Assignments

## Practical 3

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### 3.1.1 Numpy Array Operations

**Problem Statement:**
Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

**Input Format:**

- The first line contains two space-separated integers r and c, representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

**Output Format:**

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

**Code:**

```
import numpy as np
rows, cols = map(int, input().split())
elements = []

for _ in range (rows):
        elements.extend(map(int,  input().split()))

array = np.array(elements).reshape(rows, cols)
print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

## Output:

| Average time | Maximum time |
|---|---|
| **0.010 s** | **0.017 s** |
| 9.75 ms | 17.00 ms |

2 out of 2 shown test case(s) passed

10 out of 10 hidden test case(s) passed

Test case 1 17 ms Debug

| Expected output | Actual output |
|---|---|
| 3 4 | 3 4 |
| 1 2 3 4 | 1 2 3 4 |
| 5 6 7 8 | 5 6 7 8 |
| 9 10 11 12 | 9 10 11 12 |
| [[ 1  2  3  4] | [[ 1  2  3  4] |
| [ 5  6  7  8] | [ 5  6  7  8] |
| [ 9 10 11 12]] | [ 9 10 11 12]] |
| 2 | 2 |
| (3, 4) | (3, 4) |
| 12 | 12 |

Test case 2 6 ms Debug

| Expected output | Actual output |
|---|---|
| 0 0 | 0 0 |
| [] | [] |
| 2 | 2 |
| (0, 0) | (0, 0) |
| 0 | 0 |

### 3.2.1 Numpy: Matrix Operations

**Problem Statement:**
The given code takes two 3×3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

**Task:**
You are required to compute and display the results of the following matrix operations:
1. **Addition** (matrix_a + matrix_b)
2. **Subtraction** (matrix_a – matrix_b)
3. **Element-wise Multiplication** (matrix_a * matrix_b)
4. **Matrix Multiplication** (matrix_a · matrix_b)
5. **Transpose of Matrix A**

**Input Format:**
- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

**Output Format:**
The program should display the results of the operations in the following order:
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

**Code:**

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
addition_result = matrix_a + matrix_b
print("Addition (A + B):")
print(addition_result)

# Subtraction

subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(subtraction_result)

# Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)

# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a, matrix_b)
print("A dot B:")
print(matrix_multiplication_result)

# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

**Output:**

Average time **0.019 s** 18.75 ms | Maximum time **0.024 s** 24.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 24 ms Debug

| Expected output | Actual output |
|---|---|
| Enter Matrix A: | Enter Matrix A: |
| 1 2 3 | 1 2 3 |
| 4 5 6 | 4 5 6 |
| 7 8 9 | 7 8 9 |
| Enter Matrix B: | Enter Matrix B: |
| 1 1 1 | 1 1 1 |
| 1 1 1 | 1 1 1 |
| 1 1 1 | 1 1 1 |
| Addition (A + B): | Addition (A + B): |
| [[ 2  3  4] | [[ 2  3  4] |
| [ 5  6  7] | [ 5  6  7] |
| [ 8  9 10]] | [ 8  9 10]] |
| Subtraction (A - B): | Subtraction (A - B): |
| [[0 1 2] | [[0 1 2] |
| [3 4 5] | [3 4 5] |
| [6 7 8]] | [6 7 8]] |
| Element-wise Multiplication (A * B): | Element-wise Multiplication (A * B): |
| [[1 2 3] | [[1 2 3] |
| [4 5 6] | [4 5 6] |

| Average time | Maximum time |
|---|---|
| **0.019 s** | **0.024 s** |
| 18.75 ms | 24.00 ms |

**2 out of 2 shown test case(s) passed**

**2 out of 2 hidden test case(s) passed**

| | |
|---|---|
| [7 8 9]] | [7 8 9]] |
| A dot B: | A dot B: |
| [[ 6  6  6] | [[ 6  6  6] |
| [15 15 15] | [15 15 15] |
| [24 24 24]] | [24 24 24]] |
| Transpose of A: | Transpose of A: |
| [[1 4 7] | [[1 4 7] |
| [2 5 8] | [2 5 8] |
| [3 6 9]] | [3 6 9]] |

Test case 2 20 ms Debug

| Expected output | Actual output |
|---|---|
| Enter Matrix A: | Enter Matrix A: |
| 2 4 8 | 2 4 8 |
| 9 6 5 | 9 6 5 |
| 7 8 9 | 7 8 9 |
| Enter Matrix B: | Enter Matrix B: |
| 10 12 13 | 10 12 13 |
| 14 15 16 | 14 15 16 |
| 17 18 19 | 17 18 19 |
| Addition (A + B): | Addition (A + B): |

| | |
|---|---|
| Addition (A + B): | Addition (A + B): |
| [[12 16 21] | [[12 16 21] |
| [23 21 21] | [23 21 21] |
| [24 26 28]] | [24 26 28]] |
| Subtraction (A - B): | Subtraction (A - B): |
| [[ -8  -8  -5] | [[ -8  -8  -5] |
| [ -5  -9 -11] | [ -5  -9 -11] |
| [-10 -10 -10]] | [-10 -10 -10]] |
| Element-wise Multiplication (A * B): | Element-wise Multiplication (A * B): |
| [[ 20  48 104] | [[ 20  48 104] |
| [126  90  80] | [126  90  80] |
| [119 144 171]] | [119 144 171]] |
| A dot B: | A dot B: |
| [[212 228 242] | [[212 228 242] |
| [259 288 308] | [259 288 308] |
| [335 366 390]] | [335 366 390]] |
| Transpose of A: | Transpose of A: |
| [[2 9 7] | [[2 9 7] |
| [4 6 8] | [4 6 8] |
| [8 5 9]] | [8 5 9]] |

### 3.2.2 Numpy: Horizontal and Vertical Stacking of Arrays

**Problem Statement:**

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

**Input Format:**

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

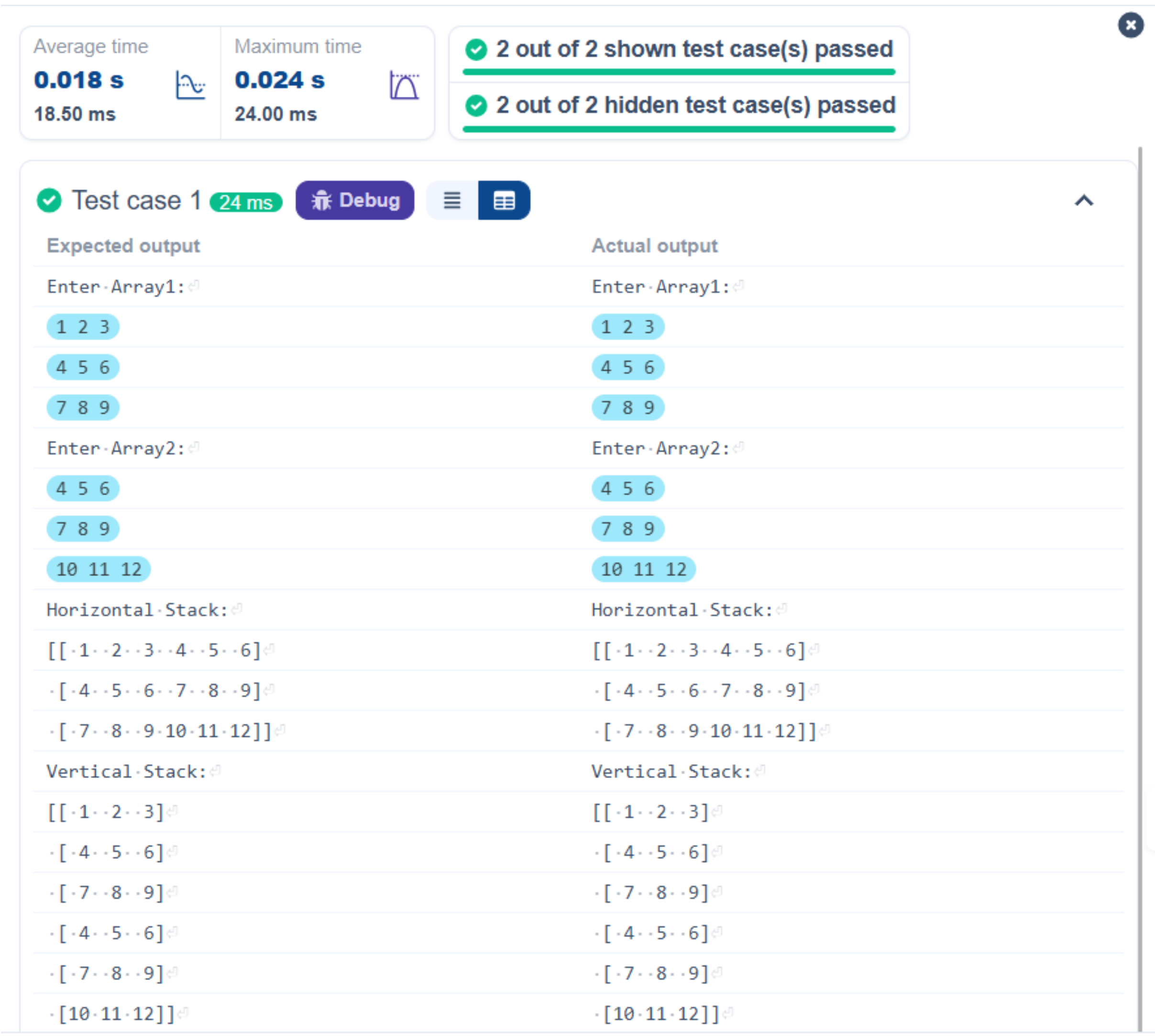
**Output Format:**

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

**Code:**

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
a = np.hstack((arr1, arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b = np.vstack((arr1, arr2))
print("Vertical Stack:")
print(b)
```

**Output:**



## 3.2.3 Numpy: Custom Sequence Generation

**Problem Statement:**

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

**Input Format:**

- The user will input three integer values: start, stop, and step, each on a new line.

**Output Format:**

- The program should print the generated sequence based on the input values.

**Code:**

```
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
a = np.arange(start, stop, step)
# Print the generated sequence
print(a)
```

**Output:**

| Average time | Maximum time |
|---|---|
| **0.009 s** | **0.015 s** |
| 9.50 ms | 15.00 ms |

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 15 ms Debug

| Expected output | Actual output |
|---|---|
| 3 | 3 |
| 10 | 10 |
| 2 | 2 |
| [3 5 7 9] | [3 5 7 9] |

Test case 2 8 ms Debug

| Expected output | Actual output |
|---|---|
| 10 | 10 |
| 20 | 20 |
| 30 | 30 |
| [10] | [10] |

### 3.2.4 Numpy: Arithmetic and Statistical Operations, Mathematical Operations and Bitwise Operators

**Problem Statement:**
You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

**1. Arithmetic Operations**:
- Compute the element-wise sum, difference, and product of the two arrays.

**2. Statistical Operations**:
- Calculate the mean, median, and standard deviation of array A.

**3. Bitwise Operations**:
- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i$ OR $B_i$).

**Input Format:**
- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

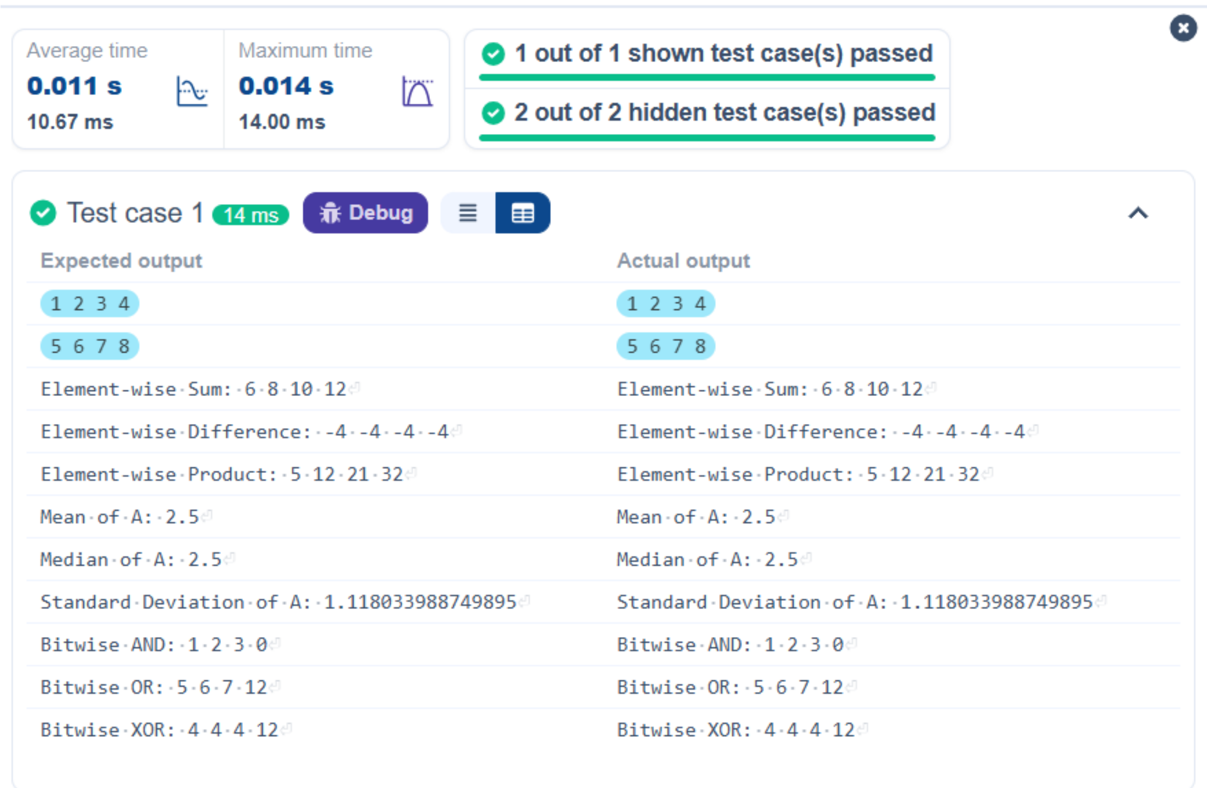
**Output Format:**
- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

**Code:**

```
import numpy as np

def array_operations(A, B):
        # Convert A and B to NumPy arrays
        A = np.array(A)
        B = np.array(B)

        # Arithmetic Operations
        sum_result = A + B
        diff_result = A - B
        prod_result = A * B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A = np.median(A)
```

```
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = A & B
        or_result = A | B
        xor_result = A ^ B

    # Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str, diff_result)))
        print("Element-wise Product:", ' '.join(map(str, prod_result)))

        print(f"Mean of A: {mean_A}")
        print(f"Median of A: {median_A}")
        print(f"Standard Deviation of A: {std_dev_A}")
          print("Bitwise AND:", ' '.join(map(str, and_result)))
        print("Bitwise OR:", ' '.join(map(str, or_result)))
        print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split()))  # Elements of array A
B = list(map(int, input().split()))  # Elements of array B
array_operations(A, B)
```

**Output:**



### 3.2.5 Numpy: Copying and Viewing Arrays

**Problem Statement:**

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

**Input Format:**

- A single line of space-separated integers.

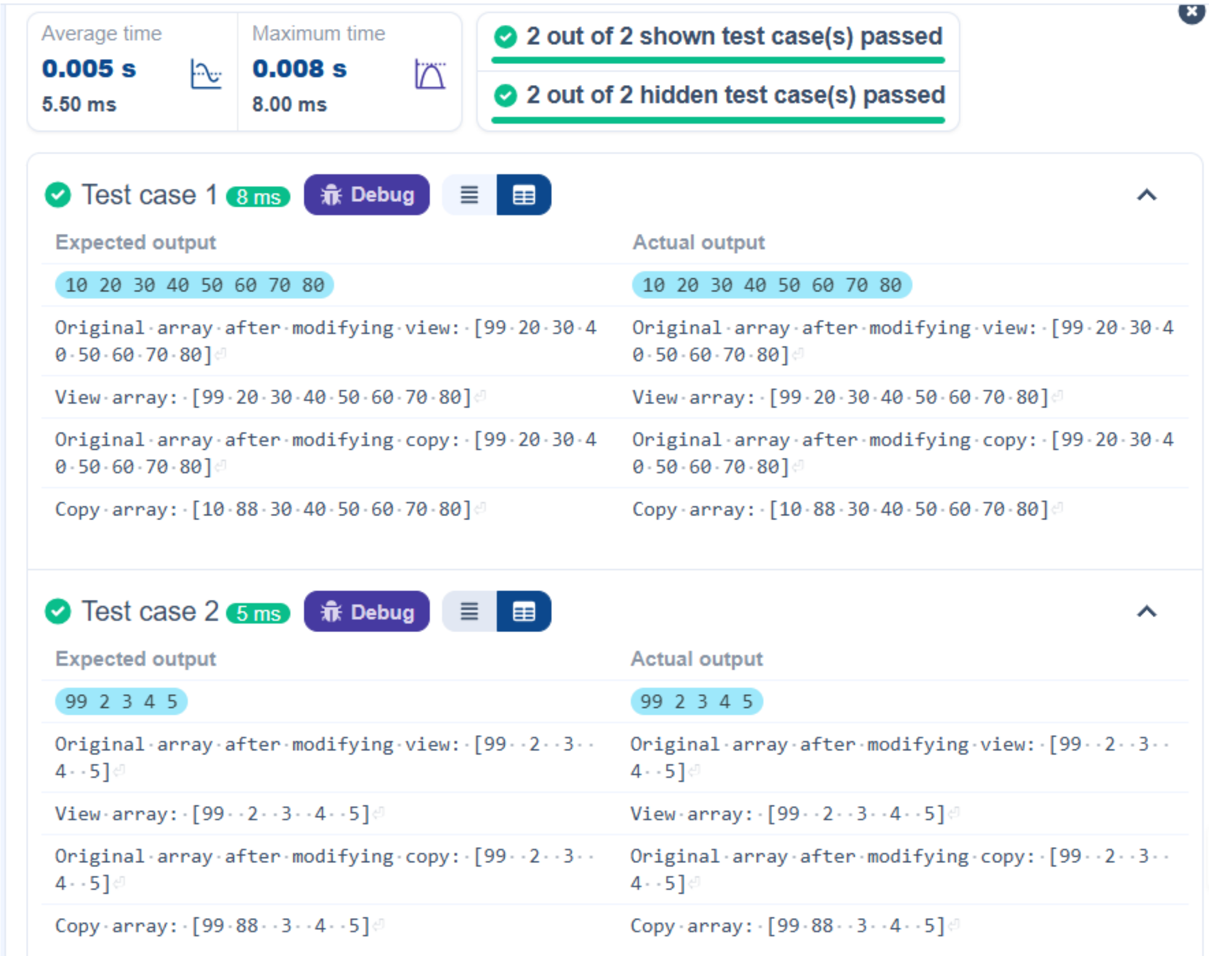
**Output Format:**

- After modifying the view:
  Original array after modifying view: <**original_array**>
  View array: <**view_array**>
- After modifying the copy:
  Original array after modifying copy: <**original_array**>
  Copy array: <**copy_array**>

**Code:**

```
import numpy as np
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

**Output:**


Average time
0.005 s
5.50 ms
Maximum time
0.008 s
8.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1 8 ms
Debug
Expected output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Actual output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Test case 2 5 ms
Debug
Expected output
99 2 3 4 5
Original array after modifying view: [99  2  3  4  5]
View array: [99  2  3  4  5]
Original array after modifying copy: [99  2  3  4  5]
Copy array: [99 88  3  4  5]
Actual output
99 2 3 4 5
Original array after modifying view: [99  2  3  4  5]
View array: [99  2  3  4  5]
Original array after modifying copy: [99  2  3  4  5]
Copy array: [99 88  3  4  5]

### 3.2.6 Searching, Sorting, Counting, Broadcasting

**Problem Statement:**

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:
**1. Searching**: Find the indices where search_value appears in array1 and print these indices.
**2. Counting**: Count how many times count_value appears in array1 and print the count.
**3. Broadcasting**: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
**4. Sorting**: Sort array1 in ascending order and print the sorted array.

**Input Format:**

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

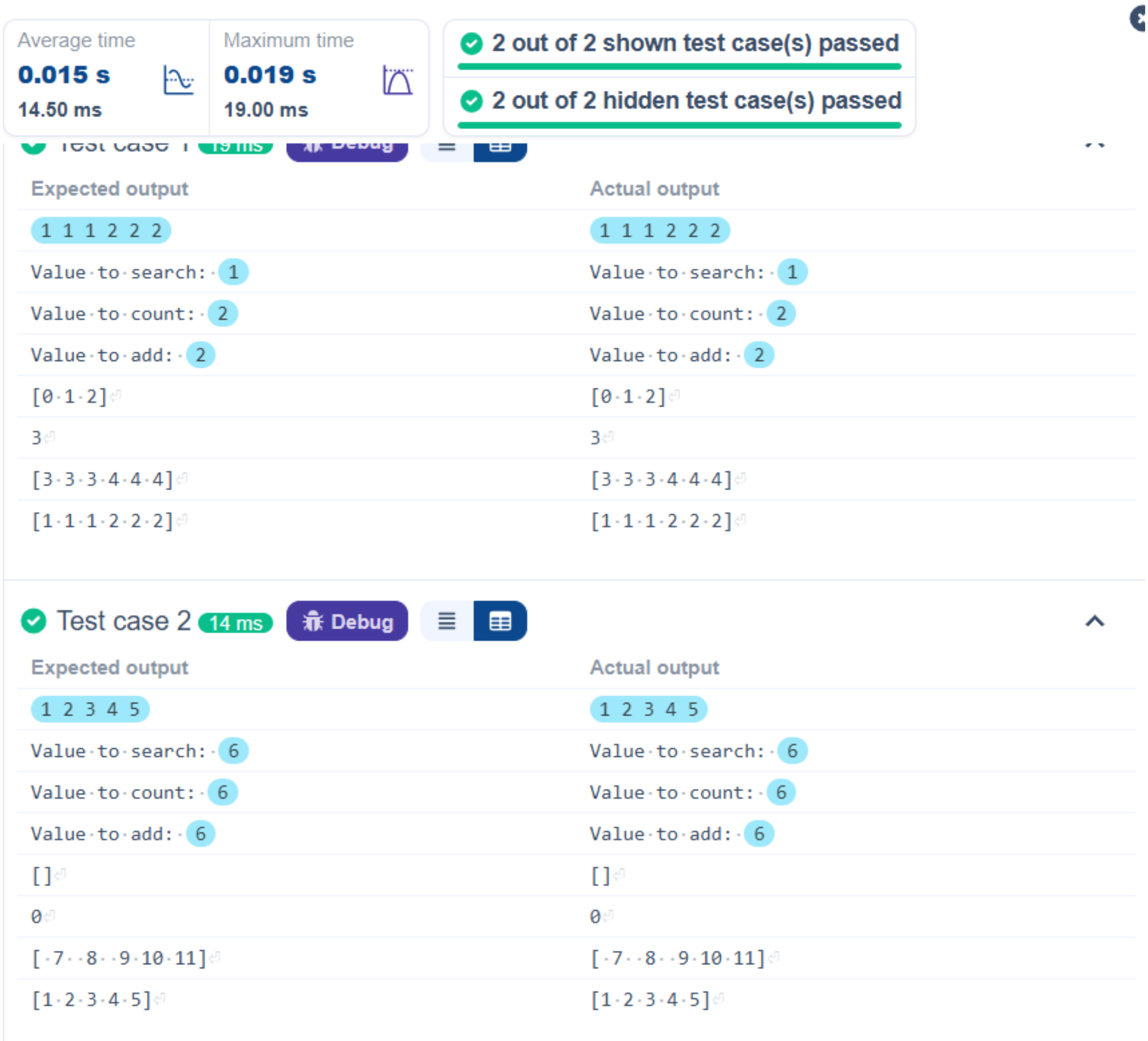
**Output Format:**

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

**Code:**

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a = np.where(array1 == search_value)[0]
print(a)

# Count occurrences in array1
b = np.count_nonzero(array1 == count_value)
print(b)

# Broadcasting addition
c = array1 + broadcast_value
print(c)

# Sort the first array
d = np.sort(array1)
print(d)
```

**Output:**




## 3.2.7 Student Data Analysis and Operations

**Problem Statement:**

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.

- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.

**Code:**

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

roll = a[:, 0].astype(float)
s1 = a[:, 1]
s2 = a[:, 2]
s3 = a[:, 3]

# 1. Print all student details
print("All student Details:\n", a)

# 2. print total students
print("Total Students:", len(a))

# 3. Print all student Roll numbers
print("All Student Roll Nos", roll)

# 4. Print subject 1 marks
print("Subject 1 Marks", s1)

# 5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(s2))

# 6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(s3))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. print Total marks of students
total = s1 + s2 + s3
print("Total Marks", total)

# 9. print average marks of each student
print(np.round(total/3, 1))

# 10. print average marks of each subject
print("Average Marks of each subject", np.mean(a[:, 1:], axis = 0))

# 11. print average marks of S1 and S2
```

```
print("Average Marks of S1 and S2", np.mean([s1, s2], axis = 1))

# 12. print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean([s1, s3], axis = 1))

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject 3", float(roll[np.argmax(s3)]))

# 14. print Roll number who got minimum marks in Subject 2
print("Roll no who got minimum marks in Subject 2", float(roll[np.argmin(s2)]))

# 15. print Roll number who got 24 marks in Subject 2
print("Roll no who got 24 marks in Subject 2",roll[s2 == 24].astype(float).reshape(1,
-1))

# 16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40", np.sum(s1 < 40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:", np.sum(s2 > 90))

# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >= 90:", np.sum(a[:, 1:] >=
90, axis = 0))

# 19. print count of subjects in which each student got marks >= 90
print("Roll no:", roll)
print("Count of subjects in which student got marks >= 90:", np.sum(a[:, 1:] >= 90,
axis = 1))

# 20. Print S1 marks in ascending order
print(np.sort(s1))

# 21. Print S1 marks >= 50 and <= 90
print(a[(s1 >= 50) & (s1 <= 90)].astype(float))
print(a.astype(float))

# 22. Print the index position of marks 79
print(np.where(s1 == 79))
```

**Output:**

| Average time | Maximum time |
|---|---|
| 0.016 s<br>16.00 ms | 0.016 s<br>16.00 ms |

1 out of 1 shown test case(s) passed

Test case 1 16 ms Debug

| Expected output | Actual output |
|---|---|
| All student Details: | All student Details: |
| [[301.  67.  77.  88.] | [[301.  67.  77.  88.] |
| [302.  78.  88.  77.] | [302.  78.  88.  77.] |
| [303.  45.  56.  89.] | [303.  45.  56.  89.] |
| [304.  88.  98.  45.] | [304.  88.  98.  45.] |
| [305.  78.  88.  99.] | [305.  78.  88.  99.] |
| [306.  68.  78.  36.] | [306.  68.  78.  36.] |
| [307.  79.  12.  45.] | [307.  79.  12.  45.] |
| [308.  46.  45.  77.] | [308.  46.  45.  77.] |
| [309.  89.  32.  88.] | [309.  89.  32.  88.] |
| [310.  79.  24.  33.]] | [310.  79.  24.  33.]] |
| Total Students: 10 | Total Students: 10 |
| All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] | All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] |
| Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.] | Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.] |
| Min marks in Subject 2 12.0 | Min marks in Subject 2 12.0 |
| Max marks in Subject 3 99.0 | Max marks in Subject 3 99.0 |
| All subject marks: [[67. 77. 88.] | All subject marks: [[67. 77. 88.] |
| [78. 88. 77.] | [78. 88. 77.] |

| Average time | Maximum time | |
|---|---|---|
| 0.016 s<br>16.00 ms | 0.016 s<br>16.00 ms | 1 out of 1 shown test case(s) passed |

| | |
|---|---|
| [45. 56. 89.] | [45. 56. 89.] |
| [88. 98. 45.] | [88. 98. 45.] |
| [78. 88. 99.] | [78. 88. 99.] |
| [68. 78. 36.] | [68. 78. 36.] |
| [79. 12. 45.] | [79. 12. 45.] |
| [46. 45. 77.] | [46. 45. 77.] |
| [89. 32. 88.] | [89. 32. 88.] |
| [79. 24. 33.]] | [79. 24. 33.]] |
| Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.] | Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.] |
| [77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3] | [77.3 81.  63.3 77.  88.3 60.7 45.3 56.  69.7 45.3] |
| Average Marks of each subject [71.7 59.8 67.7] | Average Marks of each subject [71.7 59.8 67.7] |
| Average Marks of S1 and S2 [71.7 59.8] | Average Marks of S1 and S2 [71.7 59.8] |
| Average Marks of S1 and S3 [71.7 67.7] | Average Marks of S1 and S3 [71.7 67.7] |
| Roll no who got maximum marks in Subject 3 305.0 | Roll no who got maximum marks in Subject 3 305.0 |
| Roll no who got minimum marks in Subject 2 307.0 | Roll no who got minimum marks in Subject 2 307.0 |
| Roll no who got 24 marks in Subject 2 [[310.]] | Roll no who got 24 marks in Subject 2 [[310.]] |
| Count of students who got marks in Subject 1 < 40 0 | Count of students who got marks in Subject 1 < 40 0 |
| Count of students who got marks in Subject 2 > 90: 1 | Count of students who got marks in Subject 2 > 90: 1 |

Average time

**0.016 s**

16.00 ms

Maximum time

**0.016 s**

16.00 ms

**1 out of 1 shown test case(s) passed**

| | |
|---|---|
| Count of students in each subject who got marks >= 90: [0 1 1] | Count of students in each subject who got marks >= 90: [0 1 1] |
| Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] | Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.] |
| Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0] | Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0] |
| [45. 46. 67. 68. 78. 78. 79. 79. 88. 89.] | [45. 46. 67. 68. 78. 78. 79. 79. 88. 89.] |
| [[301. 67. 77. 88.] | [[301. 67. 77. 88.] |
| [302. 78. 88. 77.] | [302. 78. 88. 77.] |
| [304. 88. 98. 45.] | [304. 88. 98. 45.] |
| [305. 78. 88. 99.] | [305. 78. 88. 99.] |
| [306. 68. 78. 36.] | [306. 68. 78. 36.] |
| [307. 79. 12. 45.] | [307. 79. 12. 45.] |
| [309. 89. 32. 88.] | [309. 89. 32. 88.] |
| [310. 79. 24. 33.]] | [310. 79. 24. 33.]] |
| [[301. 67. 77. 88.] | [[301. 67. 77. 88.] |
| [302. 78. 88. 77.] | [302. 78. 88. 77.] |
| [303. 45. 56. 89.] | [303. 45. 56. 89.] |
| [304. 88. 98. 45.] | [304. 88. 98. 45.] |
| [305. 78. 88. 99.] | [305. 78. 88. 99.] |
| [306. 68. 78. 36.] | [306. 68. 78. 36.] |
| [307. 79. 12. 45.] | [307. 79. 12. 45.] |
| [308. 46. 45. 77.] | [308. 46. 45. 77.] |